CERTIFIED TECHNICAL ARCHITECT PROGRAM
Sovereign Bank Case Study

Corporate overview

Sovereign Bank is a major Hong Kong financial institution with a global presence. It has become one of the largest players due to acquisitions in the last decade and now boasts 80,000 employees globally with main hubs in Asia, EMEA, and the Americas.

Through many years of mergers and acquisitions, Sovereign Bank has grown organically, and so has their IT infrastructure estate. They've struggled for many years using an outdated on-premise IT Service Management solution, Remedy, which they have highly customized.

From the outset of the original Remedy project, it was overwhelmed with challenges, including an 18-month phase one implementation. Then, five years ago, an upgrade took 18 months from start to finish (something they do not want to repeat!). Sovereign Bank is not seeing the desired value in the Remedy platform and is looking to move this strategically critical platform to the cloud. This move is a directive from the group CIO. Other leaders in the business units are a lot more reserved and are concerned about this approach due to the risk-averse nature of the bank encapsulated by a "this is the way we've always done things" mentality.

Finally, there is an instance of HR implemented by ServiceNow. The transformation needs to deal with HR profile data which is on an older release.

Sovereign Bank key desired outcomes

- Goal 1: A centralized best in-class CMDB
- Goal 2: A simplified omnichannel customer support desk
- Goal 3: Simplification and automation of processes across the enterprise
- Goal 4: Increased automation of controls to reduce risk
- Goal 5: Compliance with country specific financial industry regulations
- Goal 6: Ease of upgrade (ServiceNow release + 8 weeks)
- Goal 7: Extendibility of the platform across the Sovereign Bank Group
- Goal 8: Be the ServiceNow poster child for the 'best in class' out of the box implementation and to make Sovereign Bank fit the platform, not the platform fit the bank

Sovereign Bank key players

The main players in this ServiceNow program are:

- **Zhang Wei** – Project sponsor and key decision maker. New to the organization and keen to instil sweeping changes.

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- **Albert Wan** - Program director and has the ear of the CIO given his long tenure at the bank. Owns the platform, driving the business requirements through process change and looking to limit the technical debt on the platform.

- **Li Na** – Enterprise architect. Li has been with Sovereign Bank for 15 years and knows the estate inside out. She knows this will not be an easy program and will need much customization to keep the incumbent tool users happy. Given her experience in the bank, Li has good relationships with the business unit general managers.

ServiceWow Consulting Services

ServiceWow Consulting Services (SCS) have been in the consulting business for 15 years, mainly implementing ITSM products and services. For the past five years they have consolidated their consulting business around the ServiceNow platform. They have expanded their skillsets into CSM, HR and IT operations.

Suzi Matsui, a technical architect for SCS, has been assigned by Sovereign Bank as the lead architect to ensure leading practices prevail on the roll-out of the ServiceNow platform. Suzi has been recently promoted, and this will be her first large global deployment project.

You will be working closely with Suzi on the Sovereign Bank project. At times you may take the lead on presentations to project stakeholders. This is your opportunity to learn from Suzi's experience and develop your skills.

Suzi Matsui

A little more about Suzi:

Suzi makes digital transformation real on the ServiceNow platform, focusing on integrations and automation. She loves the challenge of figuring out the best way to map available technology to the main business goals of her customers.

Suzi looks to reduce risk to customer and users, provide industry accepted leading practices, advice, and guidance, and be seen as the expert on the ServiceNow platform.

Suzi's Responsibilities

- Translate business information and technical requirements into architectural blueprint proposing solutions to achieve complex business objectives
- Provide guidance on, assist in establishing, and participate in technical governance processes
- Collaborate on design and implementation of a platform operating model that improves likelihood of successful go-lives, end-user adoption
- Provide technical evaluation of strategic and operational demands against ServiceNow platform architecture, platform capabilities, and industry accepted leading practices

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Suzi's Focus Areas

- Deliver ServiceNow as the 'platform of platforms' for Sovereign Bank
- Integrate and automate multiple systems and platforms
- Understand the bank's complex requirements and matching to ServiceNow platform capabilities
- Keep the ServiceNow platform in good health, minimizing 'rogue' customizations or technical debt, allowing the bank easier upgrades

Organization overview

- The last major upgrade to Sovereign Bank's ITSM platform was five years ago and took 18 months to complete.
- **Implementation Methodology:** Sovereign Bank will be following the ServiceNow's Implementation Methodology, Now Create, for their transformation. This was complemented by running a two-day immersion event for all stakeholders of the ITSM implementation.
- **Training plans**: Sovereign Bank will re-train their internal team supporting their Remedy platform via the ServiceNow Fundamentals training course. The approach for end user training is to use Guided Tours - no formal end user training will be undertaken, with everything being driven through the platform. Sovereign Bank does not want to have instructor-led (or virtual) sessions as using the platform "should be as simple as using Uber, Amazon or eBay…"
- **Cultural changes:** Sovereign Bank has taken the approach to change their processes to fit the platform, not change the platform to match their process. This approach was widely communicated to all stakeholders and supported from the C suite down. Several of the key stakeholders have been with Sovereign Bank for more than 20 years, and there are pockets of high resistance to this change.
- **KPI & metrics solution:** The ServiceNow Inspire team were engaged early in the current pre-sales cycle with Sovereign Bank. They produced a full Business Value Assessment (BVA), which in turn built the business case to proceed with ServiceNow. Currently, Sovereign Bank has 130+ KPI reports that must be rationalized.
- **Demand Management process:** At the point of pre-sales engagement, Sovereign Bank had multiple channels for demand with very low levels of automation and little overarching visibility across the various portfolios. After the phase one ServiceNow delivery, they will adopt ServiceNow's demand process to manage all future demands for platform functionality.
- **Resistance to change:** Due to the length of time between changes in technology, it is expected there will be significant resistance to change from some areas of the ITSM community within Sovereign Bank (specifically process owners and process managers, along with senior key service desk staff). Resistance will likely manifest itself by quoting 'imagined' regulatory requirements, non-engagement in activities (citing other imperative priorities), protecting fiefdoms, and escalation to senior managers who are external to the program.
- **Expected outcomes:** There will be a significant personnel impact to Sovereign Bank staff, with staff redeployment and likely redundancies being made.

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Executive leadership communication to-date: Using their internal community forum, targeted email campaigns, and immersion days, the Sovereign Bank senior leadership team has communicated openly and frequently regarding the program and the desired outcomes it is tasked with delivering.

Sovereign Bank high-level outcomes

Sovereign Bank has been in talks with ServiceNow over the last 18 months around platform capabilities to deliver the following business outcomes. They have procured the following components of the platform:

	Area
1.	ITSM
1a.	Configuration Management Database (CMDB)
2.	ITOM
2a.	Visibility: Discovery
2b.	Visibility: Service Mapping
2c.	ITOM Optimization
3.	CSM
4.	HR
5.	ITBM
6.	Agent Intelligence
7.	Security / Regulatory compliance
8.	Integrations
9.	Reporting and Metrics

Desired business outcomes

	Area	Current State	Desired Future State
1.	ITSM – Incident Management	• Due to historical mergers, different divisions in the organization have slightly different incident management processes • Incidents are assigned with a mixture of assignment rules and manual interaction • Knowledge Articles exist for some products, but not all • Cases do not exist and everything is being logged as incidents • Hundreds of SLAs in place with the incumbent Remedy system	• A single unified/consistent incident management process • Incident assignment that leverages Agent Intelligence, Virtual Assistant, as well as Agent Workspace • Guidance from ServiceNow on how to incorporate Knowledge Articles into the incident process (e.g., to create INC, to assign to 2nd level, to resolve) • Guidance from ServiceNow on how to protect/encrypt customer Social Security Numbers (SSNs) since some cases will be converted into incidents and may contain SSN • Rationalization of the myriad of existing SLAs as there are many areas of duplication

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	Area	Current State	Desired Future State
1.	ITSM – Problem Management	• Complex problem management processes that vary by BUs across geographies • Anyone can create a Problem record in draft form; the Problem Management team reviews, triages, and manages (or rejects) • When a Problem is resolved, the Problem Management team manually resolves to Incidents	• Problem process manager has spent many years developing processes, and states must be kept in new solution

1.	ITSM – Change Management	• Similar to Problem Management, there are large Change Management teams with complex and varied processes across BUs and regions
		• Most of these processes use scripted assignment and approval rules to compensate for poor configuration management data
		• Formal CAB process is in place but needs to be firmed up due to last minute submissions, and after-CAB submissions. Support teams have issues with having the correct approvers on records
		• There are four types of Change records created in current system:
		    o Standard, where approvals are not needed
		    o Emergency, where approvals are required even before a Change record is completed
		    o Business Critical, for infrastructure changes that will bring down a major business service. These require additional lead time and approvals
		    o Normal, which covers all other changes that require approval
		• Risk Assessment is provided in a spreadsheet that must be attached to the Change record. This is required for all types of Change except Standard
		• Rescheduling of a change for any reason requires the Change record be moved back to the start of the approval process
			• Ad hoc approvals (ability to add additional approvers up to a certain point in the process) can be assigned to users and groups as needed
			• Auto-assignment of Change records to appropriate users or groups
			• Ensure changes can be scheduled at the least risky time. CAB needs alerts to other work taking place on the same CIs at the same time, or if maintenance or blackout windows exist for the CIs impacted
			• Approve/reject Change records via email, with a link to Change record
			• Use Change Templates to ensure better defined, more accurate Change Requests
			• All Change Requests require a testing plan. All Changes, where possible, must be tested in a sub-prod instance first
			• All CI updates occur via the Discovery process or via Change Management
			• All Incidents that occur due to Changes are tracked for metrics, investigation, and preventive actions

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	Area	Current State	Desired Future State
		• No system to track and investigate Incidents that occur as a result of Changes • There currently is a central control function that leads to many approvals being carried out by a small number of knowledgeable individuals	
1a.	Configuration Management Database (CMDB)	• Using a customized Remedy that holds the CIs, updated through internally developed change management system, and discovered through a variety of third-party tools	• Sovereign Bank CIO needs to know what input and guidance (and from who) is needed to implement a CMDB with ITOM • Centralized CMDB being fed by bottom up and top-down discovery tools. Sovereign Bank wants to understand the four steps to implementing a CMDB with ITOM • Only the Config Team can edit CIs • CMDB updates only via Change or Discovery • Need data certification, architectural standards, and a health dashboard • Take full advantage of the CSDM
2.	ITOM		

2a.	Visibility: Discovery	• Obtaining credentials is often a challenge, and with the introduction of patterns, a higher level of creds is required for Unix
		• Discover capabilities:
		    o SCCM for Desktops and Laptops
		    o BMC Atrium for Asset Discovery
			• Replace BMC atrium with ServiceNow Discovery and ideally SCCM in the future

2b.	Visibility: Service Mapping	• Managing the entire service portfolio through spreadsheets, home-grown databases and Remedy that lack real time updates and can be significantly out of date between updates • Impact and risk analysis processes that are part of the Change Management process are time consuming and inaccurate with the result of change backouts needing to be performed on a disproportionate number of Changes raised	• There is a need to have an accurate view of the entire portfolio of services being offered at Sovereign Bank. • Expectation that 90% of the estate can be mapped out accurately and updated automatically based on the Change Management process. • Estimated to have 421 systems or applications in their estate that need to be identified and mapped

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	Area	Current State	Desired Future State
			• There will be a number of Services provided by outdated systems or housed on hardware not compatible with the patterns offered by ServiceNow. A mechanism to have these included in the total service map is necessary • The Security team is aware of the level of credentials that are required in some cases to enable the discovery and mapping. The team will need to have a list of the credentials needed and will require to be present when the pattern and mapping definition is created and when the discovery is being kicked off
2c.	ITOM Optimization	• Enterprise cloud strategy is a work in progress, expected to be refined in coming months • CIO mandate is to be at 20% cloud, 40% private, and 40% legacy infrastructure in 3 years • Cloud offerings are currently provisioned ad-hoc • Scattered deployments on AWS and Azure with a large on-site vCenter private cloud. Architectural standards call for Ansible-based configuration across all cloud resources, but this is inconsistently applied	• Consolidated and standardized cloud offerings across the enterprise • Both public and private cloud offerings supported through an internal support team spun out of the infrastructure group
3.	CSM	• Customer types: mainly B2C requests, but also some B2B partner requests for product information • Channels of customer communication: web, phone, email, self-service portals • Currently use Incident process to log 'cases' • Multiple product catalogs to reference on the intranet • No easy way to review customer historical case and product data	• Consolidated contact center where users have one place to go for help • CSM implementation with typical Case Management process, with child tasks to service a case • All historical customer case and product data in one platform • Social media added as a communication channel with customers • Separate system for customer-facing portal and internal IT systems • Communities enabled for partners only, not customers
4.	HR	• Instance running with HR profile data which is on an older release.	• Need to collect all HR services in one system, but a phase one may

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		Asking that this be migrated and upgraded to the latest version • HR systems of record: SAP SuccessFactors HR (EMEA and Asia), and Workday (North America) • Employee base requires adherence to regional labor laws and the German Works Council • HR departments are largely unfamiliar with the concept of case management. All issues and tasks are handled via email or Incident Management • Due to historical mergers, HR departments have varied processes when it comes to Incident Management	be a limited solution including "General HR Inquiry" • Integration with both HR systems of record • HR personnel motivated, knowledgeable, and skilled in case management / service desk concepts and processes • Unified, consistent processes across the enterprise
5.	ITBM	• PPM is being used to manage ongoing delivery of ServiceNow and was used for phase one delivery of HR by another delivery team • Agile development is being deployed in delivery of Now on Now	• Use ServiceNow's Agile Development application to track user stories, sprints, releases, and work assigned to developers • Financial Management is not a roadmap item now, but may be in a future phase
6.	Agent Intelligence	• No equivalent to Agent Intelligence or Virtual Agent currently in use • English is the business language, although enterprise-wide apps need to be able to support multiple languages	• Cases and tasks will be routed in multiple languages • Re-designed categories and assignment groups • Virtual Agent engaged by customers via portal; by employees via portal and Microsoft Teams (enterprise-wide collaboration app) • Virtual Agent will need to interact in multiple languages • Some conversation topics may need to interact with external systems to check availability or cause some other action to be performed • End to end automated catalog items requested through Virtual Agent • Measure the number of requests deflected by Virtual Agent and Agent Intelligence, and the estimated cost of saving these efforts

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	Area	Current State	Desired Future State
7.	Security / Regulatory compliance	• Some degree of encryption is maintained at the database level, but it is too cumbersome and causes significant issues with integrations and reporting	• A number of PII fields will need to be encrypted across the platform and depending on the country of the user and/or customer is based • Due to the Brexit, there are now two regulations instituted in EMEA: The UK General Data Protection Regulation (GDPR) and the EU GDPR, which regulates the processing by an individual, a company or an organization of personal data relating to individuals in the EU. This has a number of implications on how, where and for how long we can store data labelled as personal • Similarly, need to abide by the FISMA Act in the U.S., the Privacy Act in Australia, as well as Personal Data Protection Act 2012 (PDPA) for Asia Pacific operations • Any configuration to the platform will need to take special account to the adherence of the GDPR, FISMA, PDPA, and Privacy Act regulations as breaches can carry hefty fines • There will be some individual fields that will also need to be encrypted depending on the country of origin and real-time decryption to certain fields available to certain level of roles working on the platform • Given the encryption requirement above, there is a need to ensure that reporting is impacted minimally, although the fields will need to retain the encryption for the fields applied to and not be converted on the report • It will be necessary to have the ability to run reports based on a specific customer request to identify all the information Sovereign Bank might hold in the system
8.	Integrations	• Internal systems within private network with no outbound network access for compliance purposes. Some applications are accessed via a MicroServices layer	• More and more applications and systems in the cloud • MID servers will need to operate in a fail-safe mode in all environments

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		• Internal network has segmentations based on region as well as application security ratings • Integrations must navigate specific IP restrictions and firewalls need to be configured to allow access. For bespoke systems classified as highly critical, additional requirements like mutual authentication exist • Due to mergers and acquisitions, there is more than one Enterprise Service Bus (ESB). There is no common landscape. A sister project has been launched to harmonize the integration landscape, with a target to use one global Tibco bus. It's expected to go-live in 12 months • A dedicated IdP is accessible from the cloud to allow SSO for employees	• Authentication: SAML 2.0 integration between the ServiceNow instances and the Sovereign Bank IdP (Identity Providers) will be used. ADFS 3.0 will be used as the IdP foundation and Microsoft PhoneFactor will be leveraged to perform two-factor authentication for users accessing the ServiceNow instance from outside of the Sovereign Bank network. The two-factor authentication will be handled by Sovereign Bank infrastructure and is independent of the ServiceNow platform. ServiceNow will simply perform a redirection to the appropriate IdP when necessary • User Metadata: The type of system of record for user data will direct the type of integration required. This will often be an LDAP based system, but it may also originate from external databases, PeopleSoft, etc. There may be more than one system of record for this data • Centralized access management: BU's who manage their user and group access centrally, regardless of the application system will do so externally to ServiceNow within Azure, Active Directory, or some other enterprise directory service • Need to integrate to Slack
9.	Reporting and Metrics	• On-premise data lake is used for majority of reporting needs • 200+ KPIs currently reported on, through a mixture of data extracts and Tableau	• Technology to allow Sovereign Bank staff to self-serve reports where possible • ServiceNow reports that deliver the same outcome as current systems to ensure consistency

Additional details

Area	Current State	Desired Future State
Regulatory requirements	Europe GDPR: places strict controls on data transferred to non-EU countries or international organizations. These are detailed in Chapter V of the Regulation. Data is allowed to be transferred only when the EU Commission has deemed that the transfer destination "ensures an adequate level of protection". There are a	• Data Centre must be UK-based • Data Access: Special requirements may lead to discussions related to data segregation and isolation across the platform based upon user authorization. There may be certain user classes who are able to access sensitive information while others may not. There may be special

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	number of other privacy acts that Sovereign Bank is required to comply with given the global nature of its business	requirements affecting how authentication occurs, which identity provider authenticates users, or assertion signing/encryption needs • Refer also to the section under Security Requirements
Release strategy	Remedy currently used for ITSM	• Implementation of ServiceNow is expected to replace all instances of this • No "big bang" release due to capacity. Looking to ServiceNow for guidance on an incremental release strategy • 5 years of historical data brought across from legacy platform, includes all incidents, changes, knowledge articles, and other attachments • No license renewal on the legacy platform. Decommissioning done by internal support team
Data migration		• Migration of tickets, tasks, knowledge articles, CMDB etc. via delimited flat files, JSON, web service push • Any data normalization/transformation included as part of the project

------ End of Case Study Overview -----

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